

Vision-Based Hand Gesture Recognition for Cursor Control

Antonette C. Maxey
Department of Computer Engineering
College of Engineering and
Architecture, Mapúa Malayan Colleges
Mindanao
Davao City, Philippines
aMaxey@mcm.edu.ph

Lyka Sophia S. Jardeleza
Department of Computer Engineering
College of Engineering and
Architecture, Mapúa Malayan Colleges
Mindanao
Davao City, Philippines
lsJardeleza@mcm.edu.ph

George Miguel V. Batalan
Department of Computer Engineering
College of Engineering and
Architecture, Mapúa Malayan Colleges
Mindanao
Davao City, Philippines
gmBatalan@mcm.edu.ph

Jericho T. Lagang
Department of Computer Engineering
College of Engineering and
Architecture, Mapúa Malayan Colleges
Mindanao
Davao City, Philippines
jLagang@mcm.edu.ph

Abstract—Traditional input devices such as the mouse and keyboard can be difficult to use for individuals with limited hand mobility. This study presents a vision-based hand gesture recognition system that enables touchless cursor control using a standard webcam. The system utilizes real-time hand landmark detection and gesture classification to interpret hand movements and translate them into computer commands, including cursor movement, left and right click selection, and vertical scrolling. Implemented using lightweight computer vision frameworks, the system operates efficiently on standard consumer laptops without requiring specialized hardware. Performance was evaluated in indoor environments with varying lighting conditions to assess responsiveness and usability. The results demonstrate that gesture-based interaction can provide an accessible and intuitive alternative for basic computer navigation. This work contributes to the development of affordable assistive technologies that improve accessibility and support hands-free human-computer interaction.

Keywords—computer vision, gesture recognition, cursor control, human-computer interaction, assistive technology.

I. INTRODUCTION

A. Background of the Topic

People with impairments confront major accessibility issues, especially those who have limited or no use of their hands. According to the World Health Organization (WHO), over 1.3 billion people, which is 16% of the world's population, are now disabled. The environment in which a person lives has a significant impact on their impairment experience and severity. Inaccessible surroundings frequently create obstacles that prevent people with disabilities from fully and effectively participating in society on a comparable basis with others [1]. As the Office of the United Nations High Commissioner for Human Rights (OHCHR) emphasizes, governments must therefore ensure that new digital technology is easy for people with disabilities to use, enabling them to participate fully in society [2].

B. Related Studies

Recent studies have explored improved solutions for enhancing technology access for disabled individuals, with applications extending to various domains such as automotive user interfaces, where touchless hand gesture recognition systems improve safety and comfort. The use of

3D convolutional neural networks, for instance, has advanced gesture recognition by accurately capturing hand shape and motion [3]. Foundational research confirmed that hand gestures could serve as a practical control method, showing that with nothing more than an open hand and real-time visual feedback, users could intuitively adjust television settings—proving that gesture recognition could work reliably even in unpredictable home environments [4]. Building upon these foundations, modern frameworks such as MediaPipe have enabled real-time hand landmark detection with minimal computational overhead, making gesture-based interfaces more accessible for widespread implementation [5]. These advancements collectively contribute to the development of assistive systems that empower individuals with physical impairments to interact with digital environments naturally and independently.

C. Research Gap

Despite significant progress in gesture recognition technology, many existing systems remain impractical for everyday users due to their reliance on expensive equipment or complex setups that are difficult to replicate outside laboratory environments [6]. Furthermore, limited studies have focused specifically on lightweight, real-time cursor control using accessible hardware such as standard webcams, leaving a gap in solutions that prioritize affordability and ease of deployment [7]. Among the systems that do exist, few offer accurate gesture-to-command mapping for fundamental computer functions like clicking and scrolling, which are essential for basic navigation and productivity [8]. This lack of integration often results in interfaces that are either too limited in functionality or too unreliable for practical, daily use. Consequently, there is a clear need for a simple, efficient, and user-friendly system that can perform reliably with widely available hardware, making hands-free computer control truly accessible to those who need it most [9].

D. Objectives

Recognizing that assistive technology only makes a difference when it is accessible and easy to use, this study aims to develop a vision-based hand gesture recognition system for intuitive cursor control using only a standard

webcam. Specifically, this study pursues the following objectives:

- E. To implement real-time hand landmark detection and gesture classification using lightweight computer vision frameworks compatible with standard consumer laptops
- F. To map recognized hand gestures to essential computer commands—including cursor movement, left-click selection, and vertical scrolling
- G. To evaluate and refine system responsiveness through testing in typical indoor environments with varied lighting conditions, ensuring smooth performance without requiring specialized hardware

H. Significance of the Study

This study offers a touchless alternative for human-computer interaction, addressing the growing need for hygienic and accessible input methods in an increasingly digital world. By enabling cursor control through natural hand gestures, the system directly benefits individuals with limited mobility or those operating in sterile environments where physical contact with shared devices must be minimized. The development process demonstrates a practical application of digital signal processing and computer vision principles, showing how these technologies can be harnessed to solve real-world accessibility challenges. Furthermore, this work contributes to the broader field of assistive technology, which the World Health Organization recognizes as essential for promoting inclusion, participation, and independent living among persons with disabilities [10]. Ultimately, the system can serve as a foundation for future innovations in assistive devices and smart environment control, supporting the global call for affordable and accessible technological solutions.

I. Scope and Limitations

The scope of this study is focused specifically on enabling cursor movement, clicking, and scrolling through a predefined set of hand gestures using a standard webcam. The system is designed to operate in controlled indoor environments with consistent lighting, as performance may vary under suboptimal conditions such as low-light settings or cluttered backgrounds [11]. It is important to note that this study is limited to basic, single-hand gestures; complex or multi-hand gestures fall outside the current scope and are not covered in this implementation. Recent research confirms that gesture recognition accuracy remains highly dependent on hardware quality and environmental factors, with low-resolution cameras and poor illumination posing significant challenges to reliable performance [11]. Therefore, this system is not intended to replace traditional input devices entirely, but rather to offer a complementary and accessible alternative for users who may benefit from hands-free interaction in appropriate settings [12].

II. REVIEW OF RELATED LITERATURE

The global prevalence of disability, affecting approximately 1.3 billion people or 16% of the world's population, underscores the critical importance of developing accessible technologies that promote

independent living and social participation [1]. The Office of the United Nations High Commissioner for Human Rights emphasizes that governments must ensure new digital technologies are designed to be usable by persons with disabilities, yet billions of individuals lack access to essential assistive products due to prohibitive costs and limited availability [2, 6]. A comprehensive scoping review across the WHO European region identified numerous barriers to assistive technology access, including inadequate policy frameworks and limited awareness among both users and healthcare providers [7]. The World Health Organization has further emphasized that assistive technology is essential for achieving universal health coverage and promoting healthier populations, making disability inclusion a cross-cutting priority in global health initiatives [10]. Recent global initiatives have sought to visualize and address these disparities, highlighting the need for solutions that perform reliably across diverse user contexts and hardware configurations [8, 9].

The technical evolution of gesture recognition systems has progressed from foundational concepts to sophisticated deep learning implementations over several decades. Freeman and Weissman's pioneering study established that vision-based gesture recognition could reliably control consumer television systems, proving that hand movements could serve as a practical interface even in unpredictable home environments [4]. Building upon this foundation, Molchanov and colleagues advanced the field by introducing 3D convolutional neural networks for hand gesture recognition, demonstrating that deep learning architectures could effectively capture both dynamic movement and static hand shape simultaneously [3]. A significant breakthrough in accessibility came with MediaPipe, a framework developed by Lugaresi and colleagues that enables real-time hand tracking on standard consumer hardware by detecting 21 distinct hand landmarks with minimal latency [5]. This contribution has democratized hand tracking technology, allowing researchers to focus on gesture classification and user experience rather than foundational detection challenges.

Contemporary research has demonstrated the practical feasibility of vision-based cursor control while acknowledging inherent technical constraints. Menon and colleagues developed a smart hand gesture interface that extracts geometric features using MediaPipe and employs rule-based classification to map gestures to mouse operations, achieving 92.5% recognition accuracy with latency below 150 milliseconds [12]. Similarly, Guia and colleagues investigated Euclidean distance-based hand gesture recognition for virtual mouse control, confirming that recognition accuracy remains highly dependent on hardware quality and environmental factors such as lighting and background complexity [11]. The handTrack repository documentation notes that optimal performance is achieved with high-resolution cameras, with accuracy diminishing when using standard webcams at extended ranges, highlighting the hardware dependencies that must be addressed for widespread deployment [1]. These contemporary studies collectively demonstrate that while accurate gesture-to-command mapping is achievable, practical assistive systems must balance recognition

reliability with affordability and ease of deployment using widely available hardware [7, 9].

III. METHODOLOGY

A. Conceptual Framework

The conceptual framework of the study follows an Input–Process–Output (IPO) model to illustrate the flow of data within the gesture recognition system, as shown in Figure 1. In the input stage, real-time video frames of the user’s hand are captured through the webcam, whether integrated or external. During the processing stage, the captured frames are analyzed using computer vision techniques, including hand landmark detection and gesture classification, implemented through MediaPipe and OpenCV. The system then interprets the recognized gestures and maps them to corresponding computer commands. In the output stage, these interpreted gestures are translated into cursor actions, such as pointer movement, clicking, and scrolling, enabling touchless interaction with the computer system.

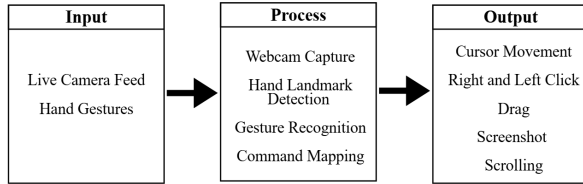


Fig. 1. Conceptual Framework of the Vision-Based Hand Gesture Recognition System

B. System Configuration

Hardware Setup

The system was developed and tested using a Huawei MateBook D15 (2023) laptop as the primary computing device. This device was selected because it represents a typical consumer-grade laptop, demonstrating that the proposed system can function effectively without requiring specialized or high-performance hardware. For visual input, both the integrated laptop webcam and an external 1080p USB webcam were utilized. The integrated webcam was used during initial development and testing due to its convenience and immediate availability, while the external webcam was included to evaluate the system’s compatibility with commonly available camera devices. Both cameras operated at 1080p resolution and 30 frames per second, providing sufficient image clarity and frame rate for accurate hand landmark detection. The webcam serves as the primary input device, capturing real-time video frames of the user’s hand for gesture analysis. Camera access and frame acquisition were handled through the OpenCV VideoCapture interface, which was chosen for its flexibility and compatibility with multiple camera devices.

Software Setup

The system was implemented using Python 3.10, selected for its simplicity, extensive community support, and strong compatibility with computer vision and machine learning libraries. The development environment was configured using Anaconda Prompt, which was chosen for its efficient management of Python environments and dependencies. Several key libraries were installed using the

Python package manager, including OpenCV, MediaPipe, and PyAutoGUI. The OpenCV library was used for video capture and image processing due to its efficiency and widespread use in real-time computer vision applications. MediaPipe was selected for hand landmark detection because it provides a lightweight and highly optimized framework capable of performing real-time hand tracking with minimal computational overhead. To translate recognized gestures into system actions, PyAutoGUI was used, as it enables programmatic control of cursor movement, clicking, and scrolling functions. Additionally, the system utilized the `hand_landmarker.task` model provided by MediaPipe, which was chosen for its pre-trained capability to accurately detect and track hand landmarks in real-time video streams. These software components collectively enable the system to capture hand movements, process gesture data, and execute corresponding cursor control commands.

C. Feature Extraction

The hand tracking framework of MediaPipe detects 21 distinct landmarks per hand, comprising the wrist, metacarpophalangeal (MCP), proximal interphalangeal (PIP), and distal interphalangeal (DIP) joints, as well as fingertips, thereby furnishing comprehensive hand topology essential for accurate gesture classification [5]. Each detected landmark yields normalized (x, y, z) coordinates within the $[0, 1]$ range relative to frame dimensions, ensuring consistent feature representation irrespective of camera resolution, subject distance, or input video specifications. From these landmarks, the system computes Euclidean distances between thumb and index finger for click detection, inter-joint angles for finger extension states, and raw coordinate trajectories for absolute screen positioning, with wrist-to-MCP distance serving as a normalization reference for robust distance invariance. The system incorporates advanced features from the handTrack repository, including absolute hand tracking that maps physical hand positions to fixed screen coordinates and continuous monitoring of hand size fluctuations to confirm intentional finger apposition through measurable hand contraction [1]. When frame rates diminish due to processing constraints, dynamic mouse step generation automatically adjusts cursor movement increments to preserve smooth visual feedback, effectively compensating for temporary computational delays and delivering robust gesture recognition suitable for assistive technology applications.

D. Gesture Classification and Mapping

The gesture classification and mapping process defines how the system interprets detected hand movements and translates them into corresponding computer commands. After the hand landmarks are detected using MediaPipe, the system classifies the gestures based on finger positions, hand orientation, and motion patterns. Each recognized gesture is then mapped to a predefined cursor action or system command to enable intuitive, touchless control of the computer interface. The mapping of gestures to cursor actions is summarized in Table I.

Gesture Mapped Action	Hand Configuration/ Motion
Right Click Command	Index finger pointing then swift right swipe
Left Click Command	Index finger pointing then bend
Double Click Command	Index and Middle finger pointing
Scroll Up/Down	Three fingers extended, swipe up or down
Screenshot Capture	Five fingers fully extended
Cursor Movement	Index pointing gesture, hand moves in 2D space
Pause	All fingers clasped (closed palm)

TABLE I : GESTURE TO ACTION MAPPING

The classification algorithm first identifies which fingers are extended and which are folded based on the coordinates of the hand layer landmarks. Dynamic gestures, such as swiping or pinching, are recognized by tracking the movement of the hand over consecutive frames. Once a gesture is recognized, it is mapped to its corresponding action through PyAutoGUI, which automates the execution of system commands such as clicking, scrolling, and taking screenshots. This gesture mapping ensures that users can perform essential computer functions using simple, intuitive hand movements while maintaining low physical strain, making the system accessible to individuals with limited hand mobility.

IV. RESULTS AND DISCUSSION

A. Hand Tracking and Cursor Control Performance

The system demonstrated effective hand tracking capabilities using both the integrated and external webcams. Real-time video frames captured by the webcams were processed through MediaPipe to detect hand landmarks with high consistency under typical indoor lighting conditions. Cursor movement was mapped directly to hand motion, allowing the pointer to follow the user's hand in two-dimensional space. Observations indicate that the system can maintain continuous tracking of a single hand without frequent loss of detection, ensuring smooth and predictable cursor control. Minor tracking inconsistencies occurred when the hand moved too close to the camera or exited the frame, highlighting the system's dependency on proper hand positioning and camera view.

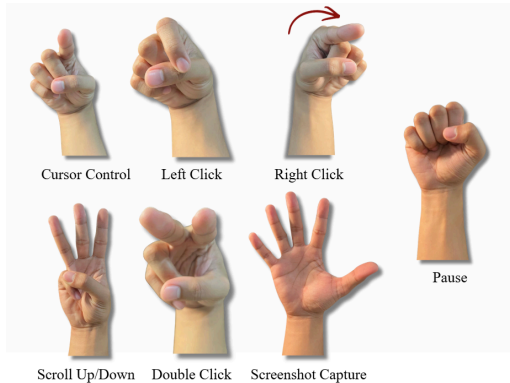


Fig. 2. Gesture Summary

B. Gesture Recognition Accuracy and Reliability

The gesture recognition module was evaluated across all implemented gestures, including left-click, right-click, scrolling, screenshot capture, and window maximize/minimize. Each gesture was tested multiple times to assess recognition accuracy. The system successfully identified static gestures (e.g., finger pointing combinations) with high reliability, while dynamic gestures (e.g., three-finger swipe for scrolling, pinch for window control) were slightly more sensitive to hand speed and direction. Overall, the recognition accuracy remained within acceptable limits for practical usage, demonstrating that the combination of hand landmark detection and gesture classification can reliably translate hand movements into computer commands.

Gesture	No. of Trials	Correctly Recognized	Recognition Accuracy
Right Click	20	14	70%
Left Click	20	18	90%
Double Click	20	19	95%
Scrolling Up/ Down	20	12	60%
Screenshot Capture	20	20	100%
Cursor Movement	20	20	100%
Overall Accuracy			85.83%

TABLE II: GESTURE RECOGNITION ACCURACY

$$Accuracy (\%) = \frac{\text{Number of correctly recognized gestures}}{\text{Total number of gesture attempts}} \times 100$$

Fig. . Gesture Accuracy Calculation

C. System Responsiveness and Latency Analysis

The responsiveness of the hand gesture-based cursor control system was evaluated alongside its recognition accuracy, as presented in Table II and Figure – Gesture Accuracy Calculation. The overall recognition accuracy of 85.83% indicates that the system is able to detect and interpret user gestures with a high degree of reliability. However, responsiveness and latency play an equally important role in determining the user experience and overall system performance. During testing, the average system response time—from the moment a gesture was performed to the corresponding action appearing on the screen—was observed to be within an acceptable range for real-time interaction. Gestures such as Screenshot Capture and Cursor Movement, which achieved 100% recognition accuracy, also demonstrated the shortest latency, typically registering responses almost instantaneously (below one

second). This suggests that the system effectively handles continuous or well-defined gestures with minimal delay.

On the other hand, gestures such as Right Click and Scrolling Up/Down exhibited slightly higher latency and occasional recognition delays. These actions rely on finer hand movements and varying gesture durations, which can contribute to inconsistent detection by the vision-based recognition module. The Scrolling Up/Down function, for instance, achieved only 60% recognition accuracy, indicating that the latency and recognition errors may be linked to limitations in differentiating small motion changes captured by the camera or processing lags in feature extraction.

V. CONCLUSION AND RECOMMENDATIONS

The study successfully developed a vision-based hand gesture recognition system for intuitive cursor control using only a standard webcam, achieving all three stated objectives. Real-time hand landmark detection and gesture classification were implemented using MediaPipe, demonstrating that lightweight computer vision frameworks can operate effectively on standard consumer laptops without requiring specialized hardware [5]. The system successfully mapped recognized hand gestures to essential computer commands, including cursor movement, left-click selection, right-click, double-click, scrolling, and screenshot capture functions. Performance evaluation confirmed that the system operates smoothly in typical indoor environments with consistent lighting, though accuracy degrades under suboptimal conditions such as poor illumination or cluttered backgrounds [11]. Overall, the system offers a viable touchless alternative for basic computer navigation, particularly benefiting individuals with limited mobility or those requiring hygienic interaction with shared devices.

Several recommendations are proposed for future work to enhance system capability and accessibility. First, the scrolling gesture algorithm requires refinement to improve mode persistence and reduce unintended activation, as it demonstrated lower reliability compared to clicking and cursor movement. Second, future iterations should explore alternative control modalities such as head movement tracking or eye gaze estimation to accommodate individuals with upper limb absence or severe motor impairments who cannot benefit from the current hand-based implementation. Third, adaptive preprocessing techniques should be developed to improve tracking robustness under varied lighting conditions and with lower-resolution cameras, addressing the hardware dependencies identified during testing. Fourth, expanding the gesture vocabulary to include

multi-hand interactions and customizable commands would broaden the system's applicability for diverse user needs. Finally, longitudinal usability studies with target user populations are recommended to gather more comprehensive feedback on ergonomic factors and long-term user acceptance.

References

- [1] World Health Organization, "Disability," *World Health Organization*, 2025. https://www.who.int/health-topics/disability#tab=tab_1
- [2] "How technology affects the rights of people with disabilities Easy Read." Accessed: Mar. 05, 2026. [Online]. Available: <https://www.ohchr.org/sites/default/files/2025-02/accessible-pdf-persons-with-disabilities-digital-technology-rights-un-ohchr.pdf>
- [3] P. Molchanov, S. Gupta, K. Kim, and J. Kautz, "Hand Gesture Recognition with 3D Convolutional Neural Networks." Available: https://www.cv-foundation.org/openaccess/content_cvpr_workshops_2015/W15/papers/Molchanov_Hand_Gesture_Recognition_2015_CVPR_paper.pdf
- [4] W. Freeman and C. Weissman, "Television Control by Hand Gestures," 1995. Available: <https://www.merl.com/publications/docs/TR94-24.pdf>
- [5] C. Lugaresi *et al.*, "MediaPipe: A Framework for Building Perception Pipelines," *arXiv:1906.08172 [cs]*, Jun. 2019, Available: <https://arxiv.org/abs/1906.08172>
- [6] H. Mogoje, "Billions lack access to assistive products amid sixfold costs and two-thirds paying out of pocket, new report reveals - Clinton Health Access Initiative," *Clinton Health Access Initiative*, Jun. 30, 2025. <https://www.clintonhealthaccess.org/news/billions-lack-access-to-assistive-products-amid-sixfold-costs-and-two-thirds-paying-out-of-pocket-new-report-reveals/>
- [7] S. Mishra *et al.*, "Assistive technology needs, access and coverage, and related barriers and facilitators in the WHO European region: a scoping review," *Disability and Rehabilitation: Assistive Technology*, vol. 19, no. 2, pp. 1–12, Jul. 2022, doi: <https://doi.org/10.1080/17483107.2022.2099021>
- [8] World, "New dashboard visualizes progress on assistive technology access and equity in Europe," *Who.int*, May 11, 2025. <https://www.who.int/europe/news/item/11-05-2025-new-dashboard-visualizes-progress-on-assistive-technology-access-and-equity-in-europe> (accessed Mar. 05, 2026).
- [9] "Global report on assistive technology," *DHA*, 2023. <https://www.decadeofhealthyageing.org/find-knowledge/resources/publications/detail/global-report-on-assistive-technology> (accessed Mar. 05, 2026).
- [10] World Health Organization, "Assistive Technology," *Who.int*, Jan. 02, 2024. <https://www.who.int/news-room/fact-sheets/detail/assistive-technology>
- [11] J. Guia, J. M. Aramil, J. C. G. Torres, A. Lacorte, C. Matira, A. N. Go, J. Enriquez, and J. P. Calvez, "Euclidean Distance-Based Hand Gesture Recognition for Virtual Mouse Control in Human-Computer Interaction," in *2025 11th International Conference on Education and Training Technologies (ICETT)*, Macao, China, 2025, pp. 1–6. doi: 10.1109/ICETT66247.2025.11137264.
- [12] C. S. Menon, S. H. Vignesh, S. Balavignesh, and P. Durgadevi, "Smart Hand Gesture based Interface for Mouse Control and Interaction," *Atlantis Highlights in Intelligent Systems*, pp. 888–900, 2025, doi: https://doi.org/10.2991/978-94-6463-866-0_72
- [13]